

Math 514, F13
Quiz 10

Name: Answer
SSN: xxx-xx-

Instructions: You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Consider an experiment with three possible outcomes and odds as follows.

Outcome	1	2	3	4
Odds	2	3	4	4

Is there a betting scheme that results in a sure win?

$$\sum_{i=1}^4 \frac{1}{1+O_i} = \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{5} = \frac{59}{60} \neq 1.$$

$\Rightarrow$ There is a betting scheme that results a sure win by the Arbitrage Theorem.

Problem 2. (5 points) Suppose that, in each period, the cost of a security either goes up by a factor of $3/2$ or goes down by a factor of $1/2$. If the initial price of the security is 100 and the nominal interest rate per period is $r = 10\%$, determine the no-arbitrage cost of a call option to purchase the security at the end of two periods for a strike price of 150.

$$K=150, t=2, S=100, u=3/2, d=1/2.$$

$$100 \begin{cases} 150 \begin{cases} 225 = S_{uu} \\ 75 = S_{ud} \end{cases} \\ 50 \begin{cases} 75 = S_{du} \\ 25 = S_{dd} \end{cases} \end{cases}$$

$\Rightarrow$ The corresponding payoffs are

$$C_{uu} = (225 - 150)(1+r)^{-2} = 75(1.1)^{-2} \approx 61.98$$

$$C_{ud} = C_{du} = C_{dd} = 0$$

$$\text{the risk-neutral probability } P = \frac{1+r-d}{u-d} = \frac{1.1-0.5}{1.5-0.5} = 0.6$$

$$\begin{aligned} \Rightarrow C &= P^2 C_{uu} + P(1-P) C_{ud} + P(1-P) C_{du} + (1-P)^2 C_{dd} \\ &= P^2 C_{uu} = 0.6^2 \times 61.98 \approx 22.31 \end{aligned}$$